# Arul Rhik Mazumder

U.S. Citizen

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**Research interests.** The cost of useful quantum computation: proving how many expensive operations (T gates) and extra qubits a circuit needs, designing cheaper algorithms for error-corrected quantum computers, and finding where quantum algorithms beat classical ones.

## Education

**Imperial College London** **Sep. 2026 – Sep. 2027 (expected)**
*MSc, Physics with Quantum Dynamics* *London, UK*

**Carnegie Mellon University** **Aug. 2023 – May 2026**
*B.S. in Computer Science; ML and Algorithms & Complexity concentrations; University Honors* *Pittsburgh, PA*

- Coursework: Quantum Computing, Complexity Theory, Learning Theory, Randomized Algorithms, Convex Optimization.

## Research Experience

**Institute for Quantum Computing (IQC), University of Waterloo** **May 2026 – Aug. 2026**
*Undergraduate Researcher (Mitacs Globalink Scholar), hosted by Michele Mosca and Vlad Gheorghiu* *Waterloo, ON, Canada*

- Proved, in a **single-author** paper (arXiv:2610.01024), that no circuit without internal Hadamard gates can implement a layer of $m$ non-overlapping Toffoli-type (CCZ) gates with fewer T gates than Campbell and Howard's $6m + 1$ construction, using a lower bound, computable in polynomial time, for a broad class of gates (diagonal Clifford+T).
- On the output of two public circuit optimizers (PyZX, TODD), the bound proves 193 of 311 circuit pieces already optimal, against 113 for stabilizer nullity, the standard prior bound.

**Independent Research** **Completed Oct. 2026**
*Single-author project, carried out at CMU, IQC Waterloo and Imperial*

- Built randomized Toffoli gates that use no extra qubits once the gate is large enough, with a T-count within a factor of eight of the fewest possible among measurement-free randomized circuits (preprint submitted).

**Magic State Labs** **Aug. 2026 – Oct. 2026**
*Research Consultant, with Pradeep Niroula* *Remote*

- Wrote criteria for checking whether a problem is a good fit for decoded quantum interferometry (DQI), a recent quantum optimization algorithm, and used them to steer an AI-agent search that found two new candidate applications (arXiv:2610.06753).

**Caltech Institute for Quantum Information and Matter (IQIM)** **May 2025 – Aug. 2025**
*SURF Undergraduate Researcher, Preskill group, with Samson Wang and James Watson (Google Quantum AI)* *Pasadena, CA*

- Developed, as first author (arXiv:2608.13862), a simulation method for early error-corrected hardware that runs many shallow random circuits and combines their results classically. For a given accuracy its circuits are exponentially shallower than standard Trotter simulation, at the cost of a modest increase in the number of runs.
- Proved that the method converges, bounded its cost, tightened the bounds for systems with weak interactions or conserved quantities, and applied them to ground-state energies.

**BlueQubit** **Oct. 2025 – Present**
*Part-time Researcher, AI and Core teams* *Los Angeles, CA (remote)*

- Benchmarked GPU-accelerated quantum simulators against commercial and open-source platforms as first author (arXiv:2607.09882); the results fed into BlueQubit's simulator cost estimators.

**Quantum Technologies Group, Carnegie Mellon University** **May 2023 – Jun. 2025**
*Researcher with Sridhar Tayur* *Pittsburgh, PA*

- Wrote a tutorial on solving QUBO problems with QAOA and quantum annealing, with five open notebooks (*Tutorials in Operations Research*, INFORMS 2025).
- Implemented semidefinite-programming formulations for entanglement detection (CVXPY, PICOS, Gurobi).

**Algorithms Group, George Mason University** **May 2022 – Jun. 2025**
*Researcher with Fei Li* *Remote*

- Three papers on quantum and hybrid algorithms: string matching in Qiskit (AISS 2022); a hybrid D-Wave algorithm for the traveling salesman problem that beat pure classical and pure quantum baselines (IPDPS 2023); and shipment rerouting with up to 1,000 shipments on real transport networks (JSSR 2024).

## Publications

### Preprints

1. **A. R. Mazumder**. "Exact $T$-counts of CCZ Layers from an Isotropy Bound." arXiv:2610.01024, 2026.
2. **A. R. Mazumder**. "Ancilla-free Randomized Toffoli Gates and the $T$-count of Boolean Oracles." arXiv, submitted 2026.

3. **A. R. Mazumder**, P. Niroula. "Discovering New Problems for Decoded Quantum Interferometry." arXiv:2610.06753, 2026.
4. **A. R. Mazumder**, J. Watson, S. Wang. "Resource-Efficient Quantum Eigenvalue Transform with Commutator Scaling." arXiv:2608.13862, 2026.
5. **A. R. Mazumder**, H. Karapeytan, M. Z. Mullath, R. Tanin, H. Gharibyan, H. Tepanyan. "Benchmarking Zero-Setup Quantum Circuit Simulators." arXiv:2607.09882, 2026.
6. F. Li, **A. R. Mazumder**, M. Zhao. "Quantum Annealing Approaches to Shipment Rerouting Problems." arXiv:2501.05624, 2025.

**Conference and journal papers**

1. **A. R. Mazumder**, E. Towe. "An End-to-End Hybrid Quantum–Classical Sampling Workflow for Discrete Markov Random Fields." *IEEE QCE*, 2026.
2. **A. R. Mazumder**, S. R. Mazumder. "Depth-Efficient Quantum Topological Data Analysis for Regime-Specific Detection of Financial Stress." *IEEE QCE*, 2026.
3. **A. R. Mazumder**, S. Tayur. "Five Starter Problems: Solving Quadratic Unconstrained Binary Optimization Models on Quantum Computers." *Tutorials in Operations Research*, INFORMS, 2025, pp. 145–183.
4. M. Zhao, **A. R. Mazumder**, F. Li. "Improved Classical and Quantum Algorithms for the Shipment Rerouting Problems." *J. Student-Scientists' Research*, 2024.
5. **A. R. Mazumder**, A. Sen, U. Sen. "Benchmarking Metaheuristic-Integrated QAOA Against Quantum Annealing." *LNNS*, 2024.
6. F. Li, **A. R. Mazumder**. "An Adaptive Hybrid Quantum Algorithm for the Metric Traveling Salesman Problem." *IPDPS*, 2023.
7. M. Gao, R. Huang, **A. R. Mazumder**, F. Li. "Comparisons of Classic and Quantum String Matching Algorithms." *AISS*, 2022.

## Awards and Honors

- **Mitacs Globalink Research Award** and **IQC Undergraduate Research Award** (2026); **Caltech SURF Fellow** (2025); **IEEE QCE Student Travel Grant** (2026).
- **Duke STAQ NSF-funded Invited Scholar** (2025): selected for a PhD-level week on quantum simulation, error correction and hardware architecture.
- **YQuantum** (2025, Yale): Yale Quantum Institute Grand Prize; **MIT iQuHACK** (2026): 2nd of 90+ teams, NVIDIA track.
- **Citadel/Correlation One Datathon** (2024): 2nd of 150 invited from 3,000+ applicants; **HackCMU** (2023): prize for an AI waste sorter piloted with CMU Dining.
- **Regeneron Science Talent Search Scholar** (2023), top 300 of 2,000+ nationally. **Olympiads:** USA(J)MO qualifier, USAPhO Bronze, USACO Gold, USNCO National Finalist.

## Leadership and Outreach

**CMU Quantum** **Sep. 2024 – May 2026**
*Founder and President (faculty advisor: Elias Towe)* *Pittsburgh, PA*

- Founded CMU's first student quantum club and grew it to 200+ members with a 10-person leadership team; raised about $15,000 in sponsorship from General Motors and Boeing.
- Wrote the introductory teaching materials for workshops with IBM, IonQ and Quantinuum; co-designed a track for the UN Future Leaders in Quantum hackathon; ran CMU's node of the Qollab Global Hackathon (12 teams); took 15 students to UMD's Quantum Leap Career Nexus.

**BlueQubit** **Oct. 2025 – Present**
*Hackathon Lead* *Remote*

- Run hackathon tracks with Duke, Yale and Stanford, mostly for first- and second-year undergraduates (30–40 students in person and hundreds online per event). Built the Duke track from scratch: students test four hidden 30- to 36-qubit circuits to find where classical simulation breaks down.
- Helped write four tutorials and five blog posts on using GPU-accelerated simulators.

## Talks and Posters

- **QRE 2026** and **UnitaryCon 2026**, Toronto — posters, "Certified T-count floors for Toffoli networks and reversible circuits."
- **TQC 2026** — poster, "Resource-Efficient Quantum Eigenvalue Transform with Commutator Scaling" (PDF).
- **Caltech IQIM SURF Seminar**, Aug. 2025, and **Quantum Leap Career Nexus**, UMD, Nov. 2025 — talks, "Early Fault-Tolerant Quantum Algorithms for Matrix Functions via Trotter Extrapolation" (slides).
- **INFORMS Annual Meeting 2025** — talk on solving QUBO models on quantum computers.

## Technical Skills

**Languages and proof tools:** Python, Julia, C++, Lean 4
**Quantum and optimization:** Qiskit, PennyLane, CUDA-Q, D-Wave Ocean; CVXPY, PICOS, Gurobi; CUDA, MPI